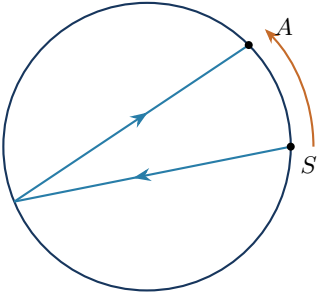
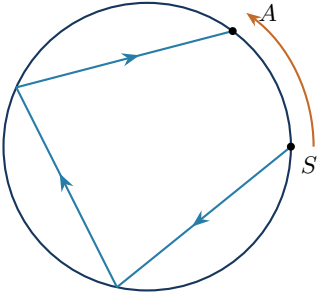


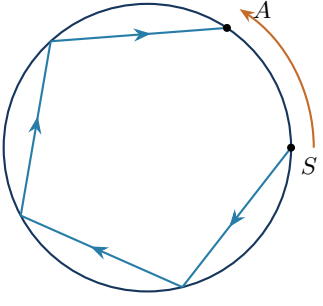
$k=2$，反射 1 次

$ct/r=3.923286$



$k=3$，反射 2 次

$ct/r=4.668407$



$k=4$，反射 3 次

$ct/r=4.920066$